

EXPERIMENT 32 — NUMBER OF VOWELS

Aim

To write and execute a Prolog program to find the **number of vowels** in a given word.

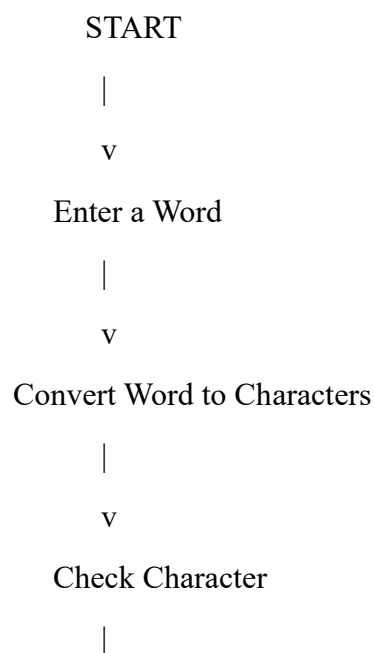
Objective

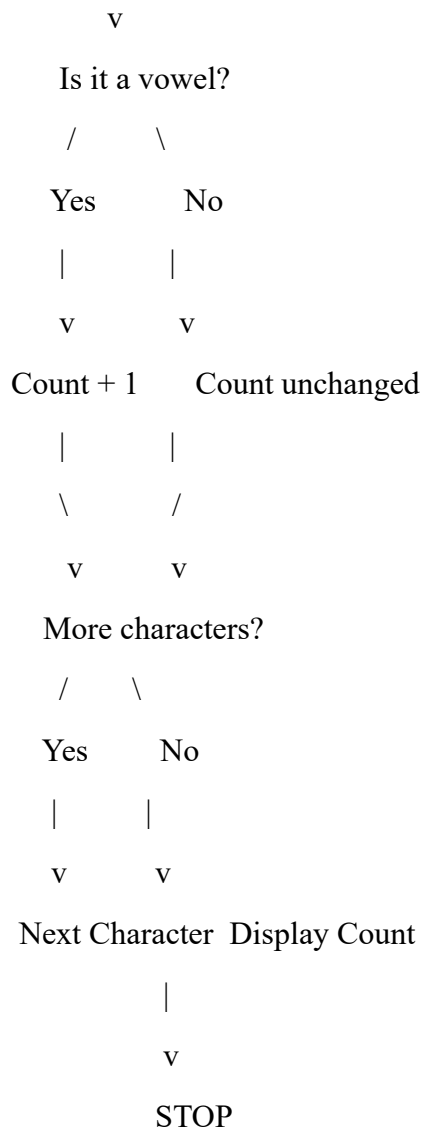
- To identify vowels in a word.
- To use recursion for processing characters.
- To count the number of vowels.
- To understand list processing in Prolog.

Algorithm

1. Start the program.
2. Define the vowels a, e, i, o, and u.
3. Convert the input word into a list of characters.
4. Check each character.
5. If the character is a vowel, increment the count.
6. If the character is not a vowel, keep the count unchanged.
7. Continue until all characters are processed.
8. Display the total number of vowels.
9. Stop.

Flowchart





Prolog Program

vowel(a).

vowel(e).

vowel(i).

vowel(o).

vowel(u).

count_vowels([], 0).

count_vowels([H|T], Count) :-

vowel(H),

```
count_vowels(T, Count1),
```

Count is Count1 + 1.

```
count_vowels([H|T], Count) :-
```

```
\+ vowel(H),
```

```
count_vowels(T, Count).
```

```
vowels_in_word(Word, Count) :-
```

```
atom_chars(Word, Characters),
```

```
count_vowels(Characters, Count).
```

Queries

Query 1

```
?- vowels_in_word(hello, Count).
```

Output

Count = 2.

Query 2

```
?- vowels_in_word(computer, Count).
```

Output

Count = 3.

Query 3

```
?- vowels_in_word(education, Count).
```

Output

Count = 5.

Sample Output

```
?- vowels_in_word(hello, Count).
```

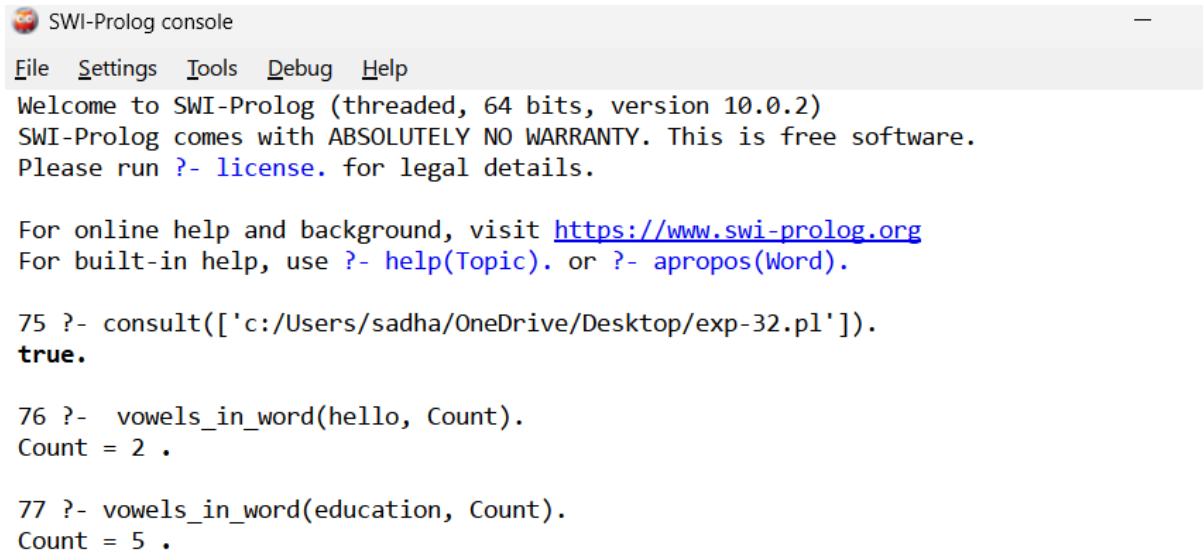
Count = 2.

```
?- vowels_in_word(computer, Count).
```

Count = 3.

?- vowels_in_word(education, Count).

Count = 5.



```
SWI-Prolog console
File Settings Tools Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

75 ?- consult(['c:/Users/sadha/OneDrive/Desktop/exp-32.pl']).
true.

76 ?- vowels_in_word(hello, Count).
Count = 2 .

77 ?- vowels_in_word(education, Count).
Count = 5 .
```

Result

Thus, the Prolog program to **find the number of vowels in a given word** was successfully implemented and executed.